# Research Methods

Day 11 · Session 11: Interpreting Findings & Discussion

# Outline for Today

1. **Draft Report Feedback**
2. **How to Write a Discussion Section: The Four-Move Structure**
3. **Team Activity: Discussion Synthesis**
4. **Synthesizing Your Findings with the Literature**
5. **Presentations: Discussion + Conclusion Slides**
6. **Scientific Storytelling: Presentation Transitions**

# Session 11: Interpreting Findings & Discussion

# Learning Objectives

### Connect findings to the question & literature

Weave your data findings into a coherent narrative that ties back to the research question and to what others have found.

### Write a Discussion that contextualizes without overclaiming

Interpret your results honestly — say what they mean, and no more than the data can support.

### Identify limitations honestly and constructively

Name the caveats that most threaten your interpretation — and what they imply.

### Begin integrating all sections into a complete presentation

Assemble intro, methods, results, discussion, and conclusion into one connected story.

# From Results to Meaning

**Results tell the audience *what* you found. The Discussion tells them *what it means*.**

The Results section reports the numbers. The Discussion is where you become the interpreter of your own evidence.

**A strong Discussion answers four questions:**

- **So what?** — why should anyone care?
- **Why?** — what might explain the finding?
- **Compared to what?** — how does it sit against the literature?
- **What are the caveats?** — where could you be wrong?

### The interpreter's job

You are the person who knows this dataset best. The Discussion is your chance to explain what the reader should take away — and to be honest about the limits of that claim.

### Discussion ≠ repeat of Results

Don't just restate every number again. Pick the findings that matter and interpret them.

# Morning: ResearchCORE

# Draft Report Feedback (20 min)

**Marked-up drafts returned**

Each RcS team gets their draft back with comments in the margins.

**Two class-wide teaching points**

We'll highlight two issues that showed up across *many* drafts — the patterns worth everyone hearing.

**10 minutes to review**

Teams read their feedback together and decide what to revise first.

**As you read your feedback, ask:**

- Which comment, if fixed, would improve the paper the *most*?
- Is any comment pointing at a problem that appears in more than one place?
- What can we fix *today* while we draft the Discussion?

10:00

# How to Write a Discussion Section

# The Four-Move Structure

Every paragraph of a good Discussion moves through the same four steps:

**1. Restate**
Say the finding in plain language — no jargon, no table numbers.

**2. Explain**
Why might this be happening? What is the mechanism?

**3. Compare**
Agree, disagree, or add nuance to what the literature said?

**4. Limit**
Name the limitation that most threatens the interpretation.

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Name the limitation that most threatens the interpretation.

**Use it as a template.** Take one key finding, walk it through all four moves, and you have a complete Discussion paragraph.

# Move 1 — Restate the Finding in Plain Language

**Plain language, not table-speak**

Translate the statistic into a sentence a non-expert would understand.

- Drop the *p*-values and coefficients for a moment.
- Say what actually happened to whom.
- Anchor it to your research question.

✕ **Instead of:**

*"The coefficient on treatment was 4.2 ($p < 0.05$)."*

✓ **Say:**

*"Students who used the study app scored about 4 points higher on the final test than those who did not."*

# Move 2 — Explain It

**Why might you have found this?** Offer a *mechanism* — a plausible story for how the cause produces the effect.

- What do we already know about *why* this relationship could exist?
- Is there a chain of steps linking your variables?
- Could something else be driving it? (name the **alternative explanation**)

**Example mechanism**

*"The app may help because it spaces practice over time, and spaced repetition is known to improve retention."*

**Alternative explanation**

*"But motivated students may have both chosen the app and studied more — so we can't rule out self-selection."*

# Move 3 — Compare to the Literature

Does your finding **agree**, **disagree**, or **add nuance** to what was already known?

### Agrees

*"This is consistent with Smith (2021), who also found a positive effect of spaced practice."*

### Disagrees

*"In contrast to Lee (2020), we found no gender difference — possibly because our sample was older."*

### Adds nuance

*"Building on Gomez (2022), we show the effect holds, but only for students who used the app weekly."*

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*"Building on Gomez (2022), we show the effect holds, but only for students who used the app weekly."*

**Every comparison needs a citation.** Tie each finding to at least one source from your literature review.

# Move 4 — Identify the Key Limitation

Every study has limits. **Name the one that most threatens your interpretation** — not a laundry list.

**Ask:**

- What would a skeptic attack first?
- Does this limitation change *what the finding means*?
- What would you do differently with more time or data?

**⚠ A limitation with teeth**

*"Because students chose whether to use the app, our estimate may reflect who they are, not what the app does. A randomized trial would settle this."*

**✓ Honest ≠ apologetic**

State the limit, say what it implies, and move on. Don't undermine your whole study.

# Interpret Without Overclaiming

✕ **Overclaiming**

- *"This proves the app causes higher scores."*
- *"Our results show this will work everywhere."*
- Treating correlation as causation.
- Generalizing beyond your sample.

✓ **Calibrated claims**

- *"Our results **suggest** the app **is associated with** higher scores."*
- *"**Within our sample**, students who…"*
- Hedging language: *may, suggests, is consistent with.*
- Scope conditions: for whom, where, when.

**The test:** could you defend every sentence of your Discussion if the author you cited were sitting in the front row?

# Team Activity:
# Discussion Synthesis

# Team Activity – Discussion Synthesis (30 min)

### Draft your Discussion

Take your **main finding** and write one paragraph using the **four-move structure**: restate → explain → compare → limit.

### Bring in the literature

Connect the finding to at least **one source** from your literature review.

### Instructor circulates

Be ready for probing questions.

**Probing questions to expect:**

- *"Why do you think you found that?"*
- *"What alternative explanation can you think of?"*
- *"Who would challenge this finding — and how would you respond?"*

30:00

# Mission Sheet: Synthesizing Findings

Before you leave the activity, your team should be able to fill in **four lines**:

1. **Main conclusion** — in one plain-language sentence, what did you find?
2. **Comparison to literature** — one source your finding agrees with, disagrees with, or extends (with citation).
3. **One important limitation** — the caveat that most threatens your interpretation.
4. **One next step** — what should the next study do to build on or fix this?

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4. **One next step** — what should the next study do to build on or fix this?

**Then:** share your four lines with a neighboring team and get quick feedback before the next segment.

# Synthesizing Your Findings with the Literature

# What Does Synthesis Mean?

**Synthesis is more than summary.** It means combining ideas from different sources to see *how they relate* — not just what each one says.

A good discussion doesn't list studies one after another. It **weaves** them together around *your* finding.

**Ask of your sources:**

- Do the authors **agree** or **disagree**?
- Are they studying the same issue from **different angles**?
- Does one source **support, expand on, or challenge** another?

**The "discuss" checklist**

To *discuss* a result means to:

1. Restate the key finding
2. Compare to what the literature said
3. Offer an explanation for it
4. Acknowledge limitations
5. Suggest what comes next

# Group Ideas, Not Authors

**✕ Don't organize by author:**

*"Smith says… then Jones says… then Lee says…"*

This reads like a list, not an argument.

**✓ Organize by theme, issue, or trend:**

*"Many studies agree that social media affects attention span (Smith, 2021; Lee, 2020), but a few challenge this idea (Gomez, 2022)."*

One sentence, three sources, one clear point.

# Use Comparative Language

### Helpful connecting phrases

- *"Similarly…"*
- *"In contrast…"*
- *"Building on…"*
- *"While some researchers argue…, others suggest…"*
- *"There is a general consensus that…"*

### Highlight gaps or conflicts

Point out where sources disagree or where more research is needed.

*"While most studies link sleep to better memory, few explore how this differs by age group."*

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*"While most studies link sleep to better memory, few explore how this differs by age group."*

**End with the big picture:** what have you learned from all the sources *combined*? What patterns, debates, or gaps set the stage for your own finding?

# Presentations: Discussion + Conclusion Slides

# Present the Ending (30 min)

Each team presents the **full ending** of their presentation — the Discussion and Conclusion slides.

**Feedback focuses on two questions:**

↔ **Does the discussion connect back?**

Do the findings tie clearly to the **original research question**?

🏁 **Does the conclusion land?**

Does it leave the audience with **one clear takeaway**?

**Draft your Discussion + Conclusion slides around three questions:**

- What do our results **mean**?
- Did they **support or challenge** our hypothesis?
- **Who should care** about this?

30:00

# Anatomy of a Strong Conclusion Slide

### One takeaway

Lead with the single sentence you want the audience to remember.

### Honest caveat

One limitation, stated plainly — it builds credibility.

### Answers the question

Explicitly close the loop back to the question you opened with.

### → Where next

A forward-looking line: what this opens up or what should follow.

**Avoid** ending on a wall of numbers or a plain "Thank you." End on *meaning*.

# Scientific Storytelling:
# Presentation Transitions

# Verbal Transitions Between Sections (10 min)

Smooth transitions signal a **well-told story**. Practice the sentences that carry your audience from one section to the next.

- **Methods → Results:** *"Now that we've seen the methods, let me walk you through what we found…"*
- **Results → Discussion:** *"Our results suggest that… This aligns with / differs from what [Author, Year] found in their study…"*
- **Discussion → Conclusion:** *"So what does all of this mean? Here is the one thing we want you to remember…"*

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**Practice out loud.** The transition is where a presentation feels either seamless or stitched together. Rehearse the hand-offs, especially when the speaker changes.

10:00

# Putting the Whole Story Together

**Question**
What did we set out to answer?

**Methods**
How did we go about answering it?

**Results**
What did we find?

**Discussion**
What does it mean, and how does it fit?

**Conclusion**
So what — and what next?

**Integration is today's goal:** every section should point back to the question and forward to the takeaway. That through-line *is* the story.

# Homework & Teamwork

# Homework — Teamwork

### As a team

- Write your **synthesis**: conclusion sentence, comparison to one literature source, one limitation, one next step.
- Draft the **Discussion slide** — *"What do our results mean?"* — connecting each finding to at least one source from your literature review.
- Finalize slide content and **assign who presents which slide**. Then **practice!**

### Slide check (async is fine)

- Are all charts labeled? Does the story flow? Identify **3 specific things to fix** in tomorrow's session.

### Individual (evening)

- Draft your **Discussion paragraph**: did the results support your hypothesis? How do findings compare to the literature? *(200–250 words)*

### Written report (RcS)

- Add a **Discussion / Conclusion** section to your written report. Include at least **2 literature comparisons with citations**. *(Target: 300–400 words)*

# Session 11: What You Can Do Now

**Interpreting findings:**

- Take a key result and move it through the **four-move structure** — restate in plain language, explain the mechanism, compare to the literature, and name the limitation that most threatens it.

**Writing the Discussion:**

- Contextualize results **without overclaiming** — use calibrated language (*suggests, is associated with*), state scope conditions, and treat correlation as correlation.
- **Synthesize** sources by theme rather than by author, using comparative language to show agreement, conflict, and gaps.

**Building the presentation:**

- Draft **Discussion and Conclusion slides** that connect findings back to the research question and leave one clear takeaway.
- Deliver smooth **verbal transitions** between sections so the whole talk reads as one coherent story.